SC3102: Signals, systems and transforms

Tutorial 4: Fourier series and continuous-time Fourier transforms

1. Determine the complex Fourier series representation for the following signals:

   (a) $x(t) = cos(4t) + sin(6t)$

   (b) $x(t) = \sum_{n=-\infty}^{\infty} y(t - 3n)$, where $y(t) = \begin{cases} e^{-t} & 0 \le t \le 3 \\ 0 & \text{otherwise} \end{cases}$

2. Compute the average power of $x(t) = e^{jt} + cos(2t + 1) - sin(5t)$

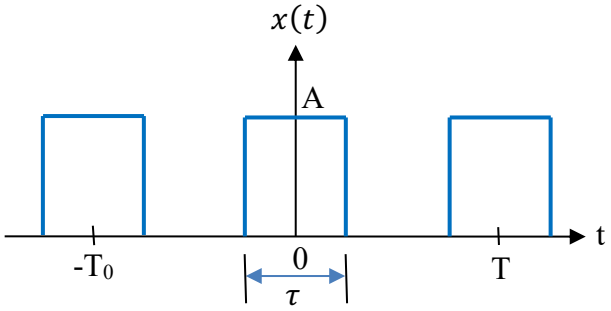
   (*Note*: use Parseval's theorem)

3. Find the Fourier transforms for the following signals:

   (a) $x(t) = cos(2\pi f_0 t)$

   (b) $x(t) = te^{-\alpha t}u(t)$, where $\alpha > 0$

4. Find the Fourier transform for the signal $x(t)$ that consists of three rectangular pulses as shown below:



5. A signal $x(t) = A + Bcos(2\pi f_0 t)$ is input to a linear time-invariant system with impulse response $h(t) = 2Wsinc(2Wt)$. Assuming $f_0 = 100\,Hz$ and $W = 1000\,Hz$, obtain the output response of the system. Sketch the power spectral density of the output signal.

   Note: $sinc(2Wt) \overset{FT}{\leftrightarrow} \frac{1}{2W}\Pi\left(\frac{f}{2W}\right)$

Answers:

1. (a) $x(t) = -\frac{1}{2j}e^{-j6t} + \frac{1}{2}e^{-j4t} + \frac{1}{2}e^{j4t} + \frac{1}{2j}e^{j6t}$

   (b) $x(t) = \frac{1-e^{-3}}{3}\sum_{k=-\infty}^{\infty} \frac{e^{j\frac{2\pi k}{3}t}}{\left(1+j\frac{2\pi k}{3}\right)}$

2. $P = \sum_{k=-\infty}^{\infty}|C_k|^2 = 4 \times \frac{1}{4} + 1 = 2$

3. (a) $X(f) = \frac{1}{2}\big(\delta(f - f_0) + \delta(f + f_0)\big)$

   (b) $X(f) = \frac{1}{(\alpha+j2\pi f)^2}$

4. $X(f) = A\tau sinc(f\tau)\big(1 + 2cos(2\pi f T_0)\big)$

5. $|Y(f)|^2 = A^2\delta(f) + \frac{B^2}{4}[\delta(f - f_0) + \delta(f + f_0)]$

Exercise:

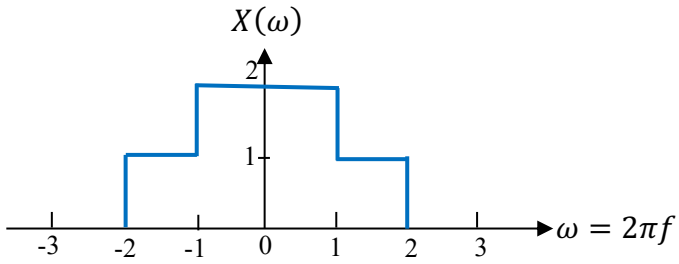
1. Determine the complex Fourier series representation for the signal $x(t) = sin^2(t)$.

2. Use the convolution theorem to evaluate the following expression.

$$\int_{-\infty}^{\infty} \frac{sin(3\tau)}{\tau}\frac{sin(t-\tau)}{t-\tau}d\tau$$

   Note: $A\Pi\left(\frac{t}{\tau}\right) \overset{FT}{\leftrightarrow} A\tau\frac{sin(\pi f\tau)}{\pi f\tau}$ where Π( ) is the rectangular function.

3. Find the inverse Fourier transform for the following function:



Answers (Exercise):

1. $x(t) = -\frac{1}{4}e^{-j2t} + \frac{1}{2} - \frac{1}{4}e^{j2t}$

2. $\int_{-\infty}^{\infty} \frac{sin(3\tau)}{\tau}\frac{sin(t-\tau)}{t-\tau}d\tau = \pi\frac{sin(t)}{t}$

3. $x(t) = \frac{1}{\pi t}\big(sin(2t) + sin(t)\big)$